

Endpoint complex Sobolev interpolation in the one-dimensional semicommutative case

Agent lane 8

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Status

This packet records a candidate partial result for arXiv:2604.23864, Hugues Moyart, *Bourgain's method for K -closedness in the semicommutative setting* [1]. The result answers the endpoint complex interpolation question for the homogeneous Sobolev couple in dimension one. It does not settle the higher-dimensional problem.

Source question

The source paper proves the non-endpoint complex interpolation identity for noncommutative homogeneous Sobolev spaces on the torus and then states that the endpoint couple is still undetermined.

For the complex method, we get the following result.

Theorem. *For every $1 \leq p_0, p_1 < \infty$ and $0 < \theta < 1$, we have*

$$[W_{1,p_0}(N), W_{1,p_1}(N)]_\theta = W_{1,p_\theta}(N)$$

with equivalent norms, with constants depending only on p_0, p_1, θ, d , where

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}.$$

The determination of complex interpolation spaces for the compatible couple $(W_{1,1}(N), W_{1,\infty}(N))$ remains open, even in the commutative case.

The relevant sentence appears on page 4 of the arXiv PDF, after the theorem on complex interpolation for $1 \leq p_0, p_1 < \infty$.

Definitions and notation

Let M be a semifinite von Neumann algebra and set

$$N = L_\infty(\mathbb{T}) \bar{\otimes} M$$

with its tensor product trace. For an exact interpolation space E between L_1 and L_∞ , Moyart defines the homogeneous Sobolev space $W_{1,E}(N)$ as the space of distributions u on $\mathbb{T}$ with $\partial u \in$

$E(N)$, equipped with the norm

$$\|u\|_{W_1, E(N)} = \|\partial u\|_{E(N)}.$$

As in the source paper, this is understood modulo the zero seminorm, i.e. modulo constant distributions. We write $W_{1,p}(N) = W_{1,L_p}(N)$.

For $E = L_1, L_\infty, L_p$, let

$$E_0(N) = \{f \in E(N) : \hat{f}(0) = 0\}.$$

Equivalently, $E_0(N) = \ker \mathbb{E}$ where

$$\mathbb{E}f = \int_{\mathbb{T}} f(z) dz$$

is the trace-preserving conditional expectation onto the constant copy of M .

Proof intuition

In one dimension there is no curl condition. The gradient range in Moyart's Leray-projection formulation is simply the zero-mean subspace of $L_p(N)$. That subspace is complemented at both endpoints by $Q = I - \mathbb{E}$. Complex interpolation is stable under a common bounded projection on the endpoints, so the endpoint Sobolev question reduces to the standard interpolation identity $[L_1(N), L_\infty(N)]_\theta = L_p(N)$.

Lemma 1 (Complemented subcouple interpolation). *Let (X_0, X_1) be a compatible Banach couple and let Q be a compatible bounded projection on both X_0 and X_1 . Put $A_j = QX_j$. Then for $0 < \theta < 1$,*

$$[A_0, A_1]_\theta = Q[X_0, X_1]_\theta$$

with equivalent norms. The constants depend only on $\|Q\|_{X_0 \rightarrow X_0}$ and $\|Q\|_{X_1 \rightarrow X_1}$.

Proof. The inclusion $[A_0, A_1]_\theta \subseteq [X_0, X_1]_\theta$ is immediate from the endpoint inclusions, and every element of the left side lies in the range of Q because the representing analytic functions have boundary values in the range of Q . Conversely, if $x = F(\theta)$ for an admissible analytic function F for the couple (X_0, X_1) , then QF is admissible for the subcouple (A_0, A_1) and $(QF)(\theta) = Qx$. Taking infima over F gives the claimed norm comparison. $\square$

Theorem 1 (One-dimensional endpoint Sobolev subcase). *Let M be a semifinite von Neumann algebra, let $N = L_\infty(\mathbb{T}) \bar{\otimes} M$, and let $0 < \theta < 1$. Define p by $1/p = 1 - \theta$. Then*

$$[W_{1,1}(N), W_{1,\infty}(N)]_\theta = W_{1,p}(N)$$

with equivalent norms.

Proof. The conditional expectation $\mathbb{E} : L_\infty(N) \rightarrow M$ is contractive and trace preserving, hence it extends contractively to $L_1(N)$. Therefore $Q = I - \mathbb{E}$ is a compatible bounded projection on the couple $(L_1(N), L_\infty(N))$, with range $(L_1^0(N), L_\infty^0(N))$ where $L_p^0(N) = \{f \in L_p(N) : \hat{f}(0) = 0\}$.

By the complemented subcouple lemma and the standard noncommutative L_p interpolation identity

$$[L_1(N), L_\infty(N)]_\theta = L_p(N),$$

we obtain

$$[L_1^0(N), L_\infty^0(N)]_\theta = L_p^0(N)$$

with equivalent norms.

It remains to identify this zero-mean couple with the Sobolev couple. In dimension one, Moyart's description of the range of the derivative says exactly that $\partial W_{1,E}(N) = E_0(N)$ for $E = L_1, L_\infty, L_p$: the higher-dimensional curl equations are vacuous and only the condition $\widehat{f}(0) = 0$ remains. Equivalently, given $f \in E_0(N)$ one defines a distribution u by

$$\widehat{u}(n) = \frac{\widehat{f}(n)}{in} \quad (n \neq 0), \quad \widehat{u}(0) = 0,$$

and then $\partial u = f$. The map $u \mapsto \partial u$ is an isometric identification of $W_{1,E}(N)$ with $E_0(N)$ by definition of the homogeneous Sobolev norm.

Transporting the previous interpolation identity through this common isomorphism of endpoint couples gives

$$[W_{1,1}(N), W_{1,\infty}(N)]_\theta = W_{1,p}(N),$$

as claimed. □

Limitations

The argument is specific to $d = 1$. For $d \geq 2$, the gradient range is the subspace of vector fields satisfying the curl-free equations and zero Fourier mean. The natural projection onto that range is the Leray projection, a matrix of Riesz-type transforms. The simple complemented-subcouple proof above would require boundedness of that projection on both L_1 and L_∞ , which is precisely the endpoint obstruction. Thus this packet should be read as a one-dimensional partial result and not as a solution of the full endpoint problem stated in the source paper.

Verification and novelty notes

The proof uses only:

- Moyart's derivative-range description for $W_{1,E}$ and $H_E(P^\perp)$;
- the standard complex interpolation identity for noncommutative L_p spaces;
- the elementary projection lemma for complex interpolation.

Local run indexes were searched for arXiv:2604.23864 and for nearby terms such as endpoint Sobolev interpolation, zero-mean subspace, and Leray projection; no prior packet for this target was found. A bounded web search for the exact endpoint formulation found no packet-level match. Because the reduction is elementary, the novelty confidence is moderate: the result may be folklore or implicit in standard one-dimensional Sobolev interpolation arguments.

References

- [1] H. Moyart, *Bourgain's method for K -closedness in the semicommutative setting*, arXiv:2604.23864, 2026.
- [2] J. Bergh and J. Lofstrom, *Interpolation Spaces. An Introduction*, Springer, 1976.
- [3] G. Pisier and Q. Xu, *Non-commutative L^p -spaces*, in Handbook of the Geometry of Banach Spaces, Vol. 2, North-Holland, 2003.